

OLDS AGDM, AB, Canada

WMO: 712580

Lat: 51.7586N

Lon: 114.0847W

Elev: 3432

StdP: 12.96

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.3	-13.8	-25.2	1.5	-19.0	-19.5	2.1	-13.4	24.7	21.7	22.0	21.8	7.9	120	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	81.8	62.1	78.3	60.6	75.2	59.1	65.5	76.6	63.3	74.1	61.2	71.6	9.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	91.9	71.7	59.0	84.6	69.1	56.8	78.1	66.6	32.4	76.7	30.5	74.7	28.9	71.6	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.1	20.8	18.4	DB	-26.3	88.2	5.2	3.8	-30.1	90.9	-33.1	93.1	-36.1	95.2	-39.9	98.0
			WB	-26.6	70.2	5.2	2.8	-30.3	72.2	-33.3	73.9	-36.2	75.5	-40.0	77.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	18.8	16.8	26.4	38.2	49.1	55.8	61.0	59.0	51.1	40.0	26.4	17.3
	DBStd	19.29	15.28	14.13	13.76	9.19	7.63	5.36	4.99	5.76	8.35	9.54	12.93	14.05
	HDD50	5251	966	929	733	363	109	10	1	5	84	328	710	1013
	HDD65	9725	1431	1349	1197	804	493	278	144	198	419	775	1159	1478
	CDD50	1039	0	0	1	9	81	185	341	284	118	19	1	0
	CDD65	38	0	0	0	0	1	3	19	12	3	0	0	0
	CDH74	935	0	0	0	3	55	81	348	302	140	6	0	0
	CDH80	181	0	0	0	0	4	8	68	69	32	0	0	0
	WSAvg	9.3	10.0	9.1	9.5	10.1	10.1	9.4	8.4	8.0	8.8	9.4	9.6	9.6
	PrecAvg	19.6	0.8	0.6	0.9	1.1	2.5	3.8	3.2	2.8	1.8	0.8	0.7	0.8
(15) Precipitation	PrecMax	24.8	1.6	1.5	1.8	3.9	5.0	7.1	8.4	4.5	4.1	1.7	1.5	1.9
	PrecMin	12.4	0.1	0.0	0.1	0.2	0.6	0.8	1.1	0.6	0.2	0.2	0.2	0.2
	PrecStd	2.9	0.4	0.3	0.4	0.7	1.1	1.5	1.5	1.1	1.0	0.4	0.4	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	47.6	58.7	71.4	79.3	80.5	85.6	86.4	84.6	73.5	60.0	48.4
		MCWB	38.0	38.0	44.7	49.8	54.6	60.5	65.0	61.1	58.0	51.3	45.5	37.5
	2%	DB	43.3	42.2	51.1	65.2	74.6	76.3	81.6	81.1	78.0	65.6	52.2	42.9
		MCWB	36.0	35.1	40.3	47.5	53.1	59.0	65.3	61.1	56.9	49.5	41.3	36.0
	5%	DB	41.1	38.3	46.3	59.1	70.1	72.5	77.9	77.0	72.7	59.8	46.4	39.4
		MCWB	35.0	32.6	38.0	44.0	51.4	57.5	63.5	61.0	55.1	46.7	37.6	33.2
	10%	DB	38.2	34.6	42.4	53.8	65.5	69.0	74.3	73.2	67.1	54.8	42.0	36.0
		MCWB	32.8	30.0	35.6	41.5	49.4	55.4	61.7	59.6	52.8	44.1	35.2	30.7
	0.4%	WB	38.7	39.0	45.8	51.0	56.7	64.9	69.8	66.6	60.3	53.9	46.2	39.1
		MCDB	46.1	45.9	57.4	69.2	74.6	75.7	79.1	77.3	77.8	70.4	59.1	45.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.8	35.4	41.2	48.1	54.2	61.4	66.8	64.3	58.1	50.3	42.0	36.1
		MCDB	42.7	41.5	49.2	63.8	71.0	72.0	78.0	75.4	74.6	64.0	51.3	42.7
	5%	WB	35.1	32.9	38.3	45.0	52.3	58.9	64.5	62.1	55.9	47.3	38.7	33.6
		MCDB	40.6	37.9	45.7	57.4	67.5	69.6	75.7	73.6	70.1	58.3	45.5	38.9
	10%	WB	32.8	30.3	35.7	42.1	50.4	56.7	62.5	60.1	53.6	44.6	35.6	30.9
		MCDB	38.1	34.1	41.9	52.3	63.4	66.8	72.5	71.2	66.3	54.0	41.0	35.8
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	17.6	17.8	20.5	22.7	20.6	22.6	23.4	22.8	21.1	17.6	16.3
		MCDBR	19.2	21.0	22.7	31.2	31.6	27.6	29.0	31.6	33.6	29.4	23.4	19.2
		MCWBR	13.8	15.1	14.3	16.5	14.3	13.4	15.4	15.2	15.5	15.8	14.5	13.9
	5% WB	MCDBR	18.8	20.2	21.9	28.9	27.5	24.3	27.1	27.7	30.6	28.3	22.8	18.7
		MCWBR	13.9	14.7	14.4	15.8	13.2	12.9	15.4	15.0	14.7	15.6	14.7	13.7
(40) Clear-Sky Solar Irradiance	taub		0.253	0.263	0.282	0.330	0.338	0.343	0.342	0.355	0.308	0.277	0.249	0.238
	taud		2.378	2.438	2.465	2.346	2.385	2.401	2.411	2.378	2.474	2.548	2.493	2.369
	Ebn at Noon		248	280	294	288	290	289	287	277	281	272	252	236
	Edh at Noon		18	23	28	35	36	36	35	34	27	21	16	16
(44) All-Sky Solar Radiation	RadAvg		346	657	1082	1468	1748	1854	2002	1646	1159	708	377	269
	RadStd		23	49	76	115	128	132	118	114	129	75	28	21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page